Linear Programming

Challenges of this module

* Generalize the optimization approaches seen for network flows
* Characterization of the optimal solution: duality theory
* Geometric aspects of Linear Programming
* Solution algorithm: improving a feasible solution
* can we give a value to resources?
* can we modify the input data and say what happens?

Link with previous knowledge

* Basic elements of linear algebra
  * matrix notation
  * solution of small systems of inequalities
  * geometric interpretation of linear functions
  * the concept of gradient (of a linear function)
* recall the "dual" vision of optimization problems

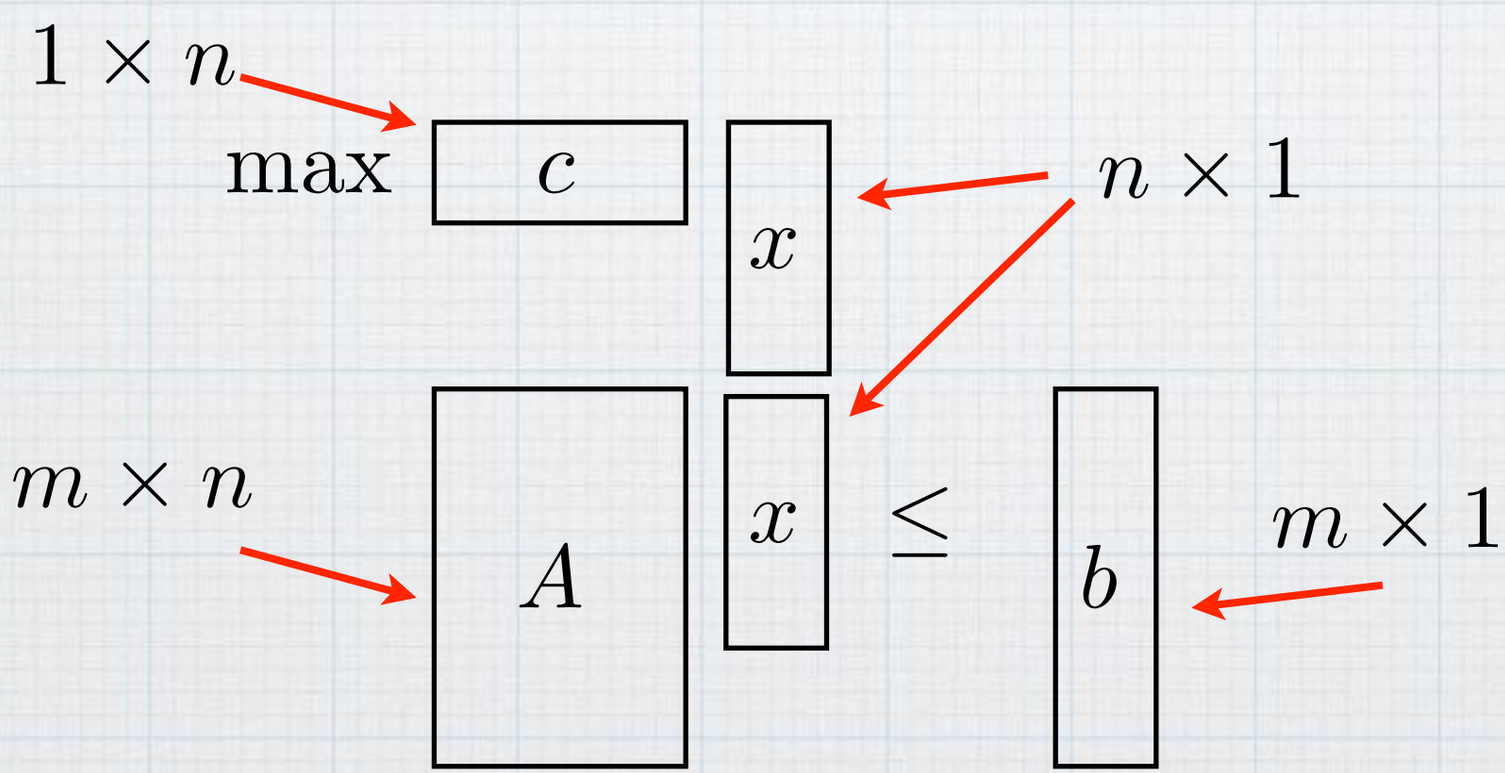
Linear optimization problems

$$\begin{aligned} &\max f(x) \\ &\quad g_i(x) \leq b_i \qquad i = 1, \ldots, m \\ &\quad x_j \in \mathcal{R} \qquad j = 1, \ldots, n \end{aligned}$$

Optimization problem

When $g_i(x)$ **and** $f(x)$ **are linear functions**

Linear Programming Problem



Doors and windows

A small firm produces:

	Aluminum doors (D)	Wooden windows (W)	
Benefit € (per item)	30	50	c

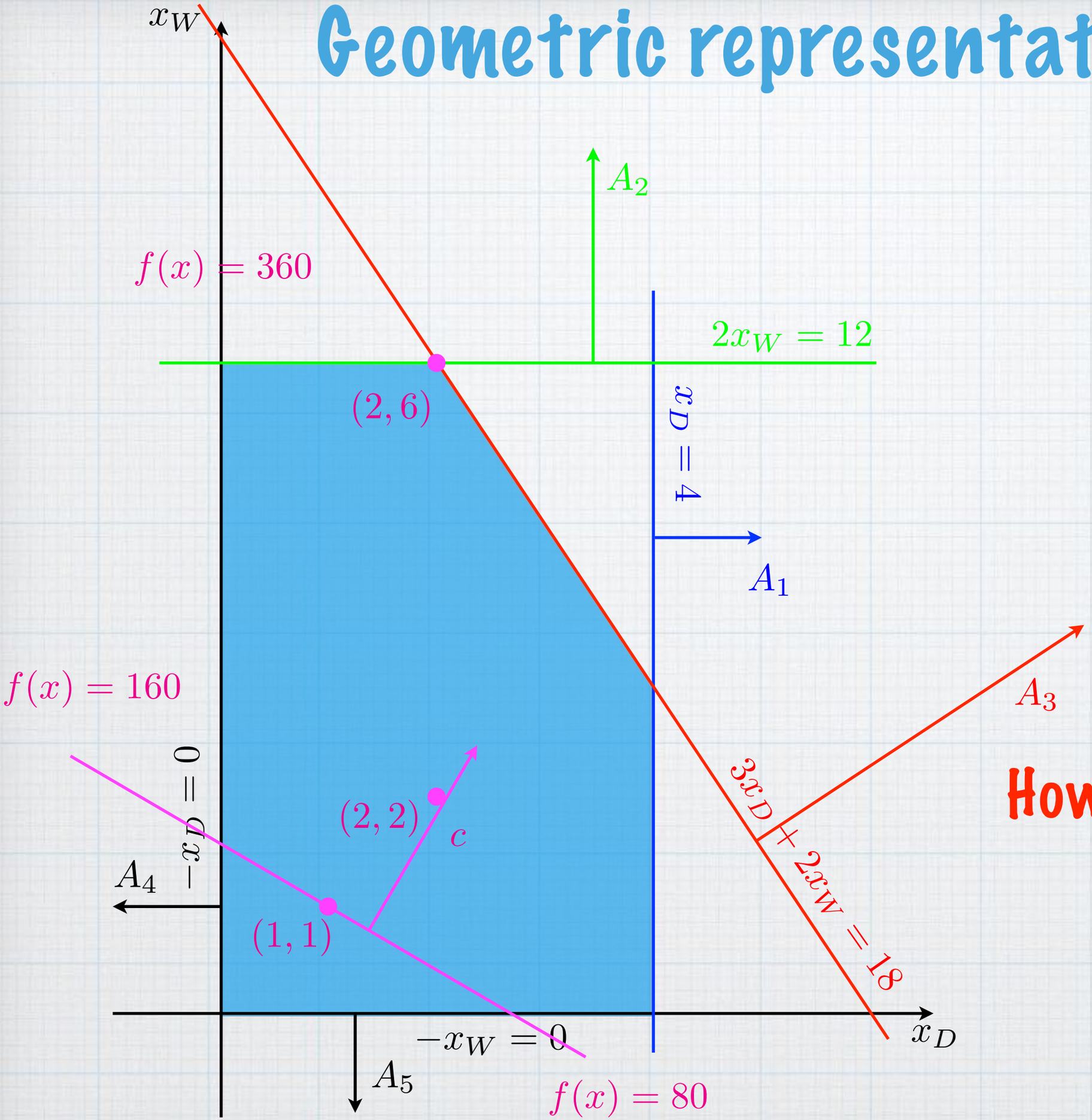
	Resources (hours per item)		Availability	
Smith	1	0	4	
Carpenter	0	2	12	b
Assembler	3	2	18	

A

Plan the production so as to maximize the return

Variables: $\begin{bmatrix} x_D \\ x_W \end{bmatrix}$ x

Geometric representation



Vertices are particularly important

How to characterize them?

Basic concepts

Convex set $S \subset \mathcal{R}^n$

convex combination

For each $x, y \in S \quad \forall \lambda \in [0,1] \quad \lambda x + (1-\lambda) y \in S$

Convex polyhedron $P \subset \mathcal{R}^n$

$$P = \{x : A_i x \leq b_i, i = 1, \ldots, m\}$$

Cone $C \subset \mathcal{R}^n$

If $x \in C$ then $\lambda x \in C, \forall \lambda \geq 0$

non negative linear combination

Convex cone $C \subset \mathcal{R}^n$

$x, y \in C \quad \forall \lambda, \mu \geq 0 \quad \lambda x + \mu y \in C$

Polyhedral cone $C \subset \mathcal{R}^n$

$$C = \{x : A_i x \leq 0, i = 1, \ldots, m\}$$

Finitely generated cone $C \subset \mathcal{R}^n$

Given $x^1, \ldots, x^r \in \mathcal{R}^n$ **rays**

$$C = \{x : x = \lambda_1 x^1 + \lambda_2 x^2 + \ldots + \lambda_r x^r, \lambda_i \geq 0, i = 1, \ldots, r\}$$

Extreme point - Vertex

Consider $P = \{x : A_i x \leq b_i, i = 1, \ldots, m\}$

$x \in P$ is an **extreme point** of P

if there are no $y, z \in P \quad y, z \neq x$

$x = \lambda y + (1 - \lambda) z$ for some $\lambda \in [0, 1]$

$x \in P$ is a **vertex** of P

if $\exists c \in \mathcal{R}^n$ such that $cx > cy, \forall y \in P, y \neq x$

The two definitions are equivalent

Basic solution

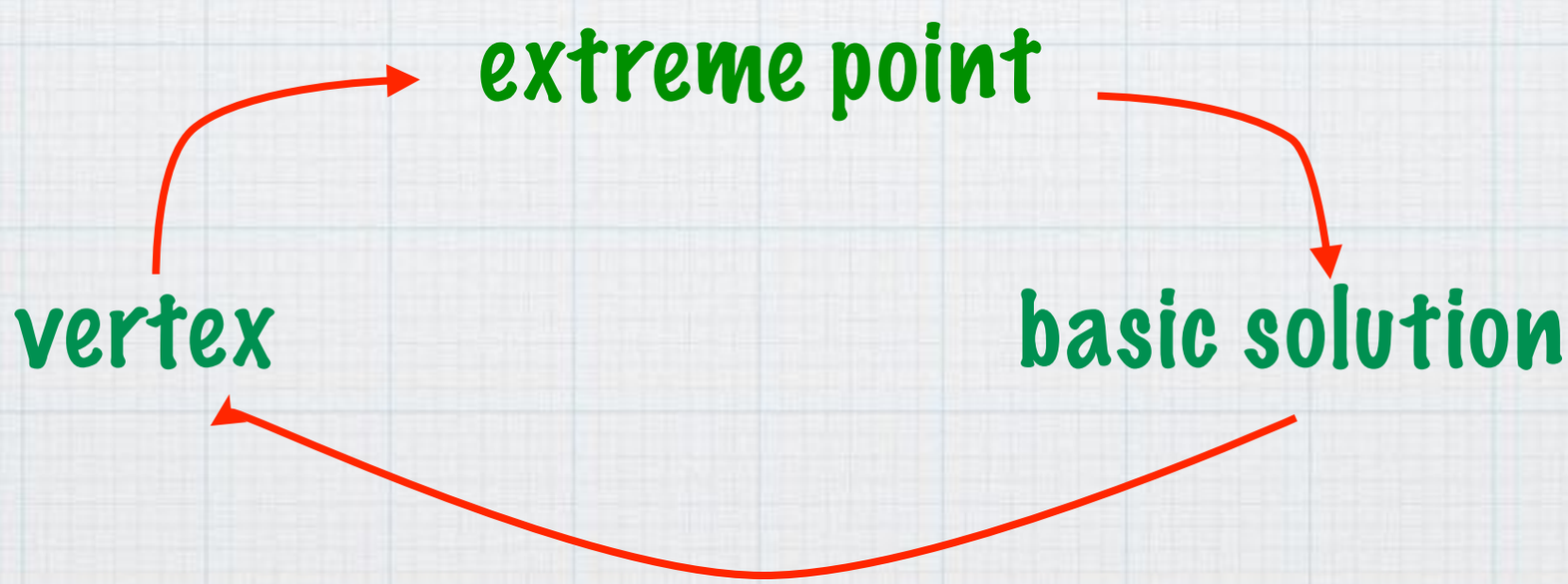
Consider $P = \{x : A_i x \leq b_i, i = 1, \ldots, m\}$

$x^* \in P$ $\quad I(x^*) = \{i : A_i x^* = b_i\}$

Set of active constraints

x^* is a **basic solution** if among all $A_i : i \in I(x^*)$

there are n of them **linearly independent**



Some properties

Consider $P = \{x : A_i x \leq b_i, i = 1, \ldots, m\}$

$$P = Q + C$$

$Q = conv\{x^1, \ldots, x^s\}$ polytope

$C = cone\{y^1, \ldots, y^t\}$ cone

If $\max\{cx : x \in P\}$ has finite optimum

then one of the extreme points of Q is optimum

Given a convex set $S \subset \mathcal{R}^n$ and a point $c \notin S$

there is a separating hyperplane

$$x \in \mathcal{R}^n : cx > zx \qquad \forall z \in S$$

Two problems

	GDA	feed A	feed B
carbohydrates	14	2	2
proteins	20	4	2
vitamins	9	1	3
cost		12	16

Farmer: minimize the cost for the feeds

Pill seller: maximize the profit

Pair of dual problems (symmetric)

P:

$$\min cx$$
$$Ax \geq b$$
$$x \geq 0$$

Farmer

D:

$$\max yb$$
$$yA \leq c$$
$$y \geq 0$$

Chicken pill seller

Let $\bar{x}$ and $\bar{y}$ be feasible solutions for P and D

then $c\bar{x} \geq \bar{y}b$

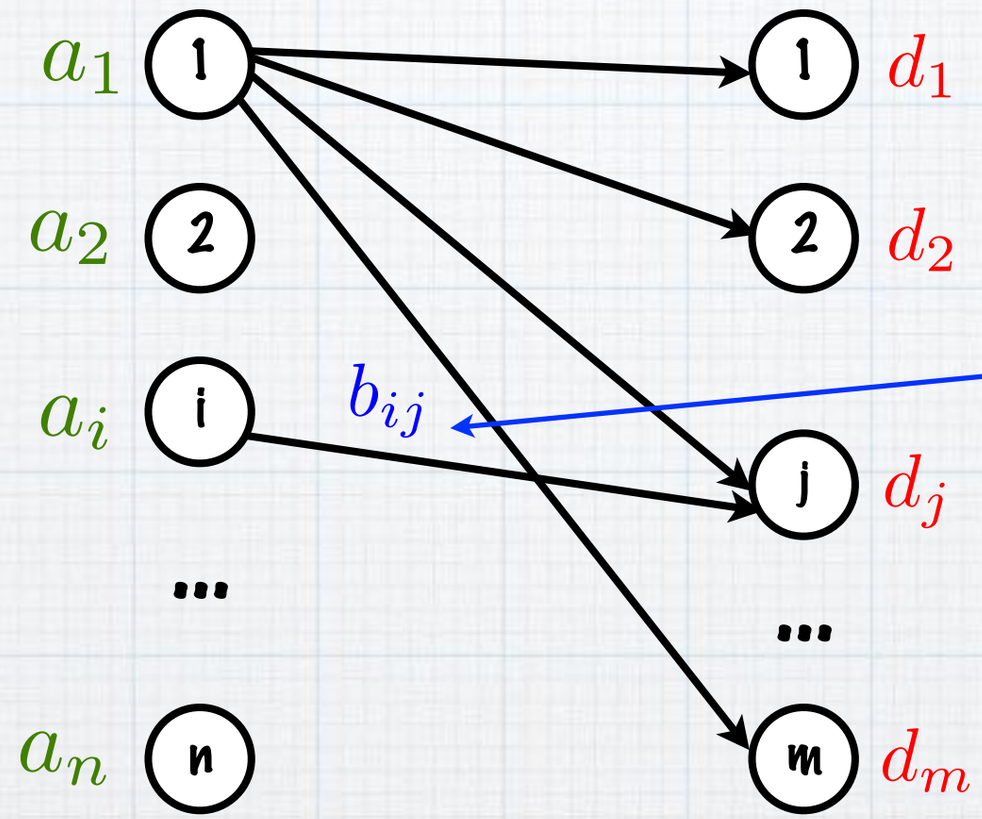
Important consequence if = holds

Other two problems

Factories Stores

production

$$\sum_{i=1}^{n} a_i = \sum_{j=1}^{m} d_j$$



request

transportation cost

Owner of factories and stores: transport the production

minimize the cost

Logistic company: buy from the factories and resell to the stores

maximize the profit

Pair of dual problems (asymmetric)

D:

$$\min yb$$
$$yA = c$$
$$y \geq 0$$

Owner of factories and stores

P:

$$\max cx$$
$$Ax \leq b$$

Transporter

Let $\bar{x}$ and $\bar{y}$ be feasible solutions for P and D

then $c\bar{x} \leq \bar{y}b$

Important consequence if = holds

Writing the dual

min	max
variables	constraints
constraints	variables
cost vector	right hand side
right hand side	cost vector
$A_i x \geq b_i$	$y_i \geq 0$
$A_i x \leq b_i$	$y_i \leq 0$
$A_i x = b_i$	y_i unrestricted
$x_i \geq 0$	$yA^i \leq c_i$
$x_i \leq 0$	$yA^i \geq c_i$
x_i unrestricted	$yA^i = c_i$

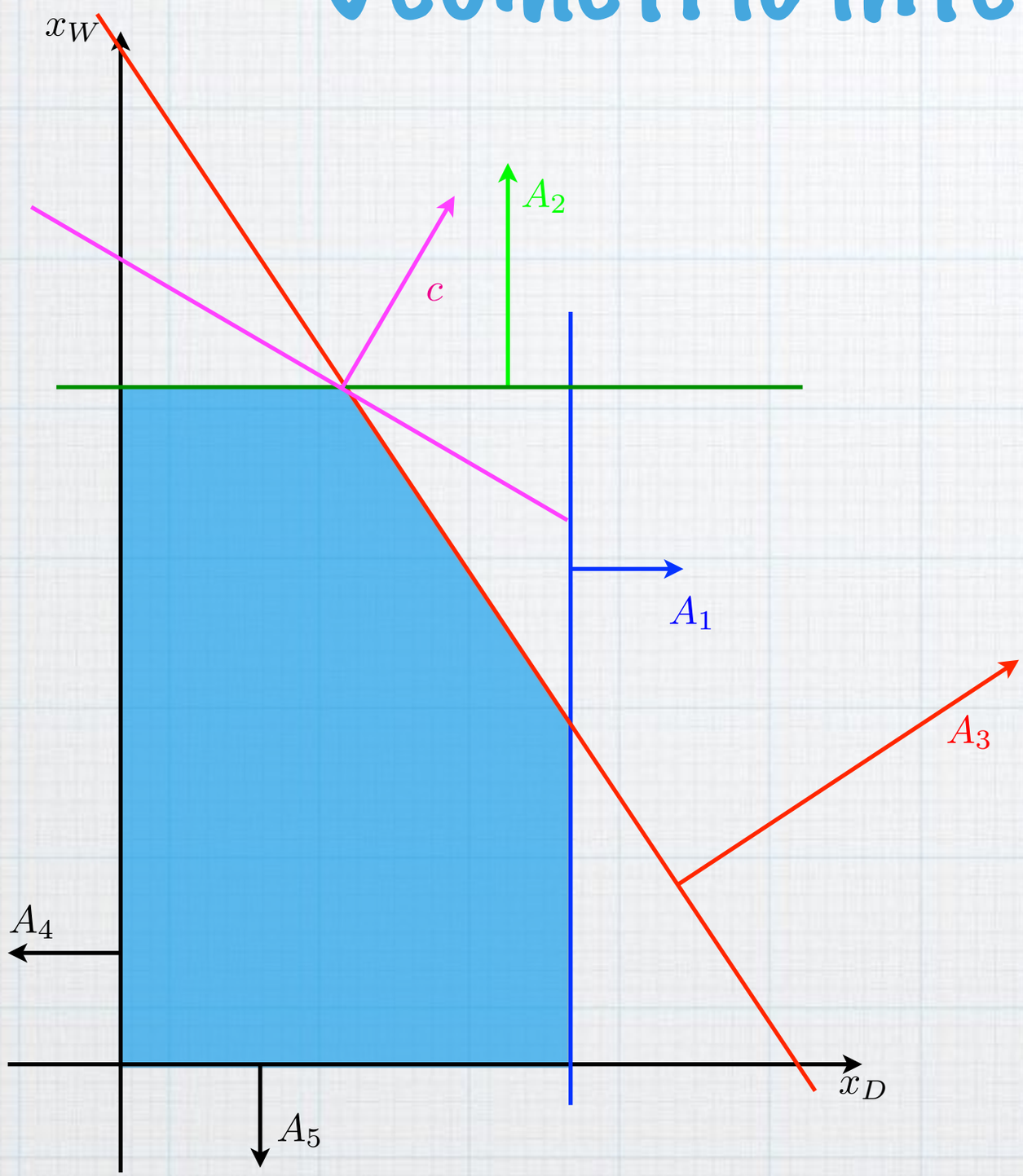
Recall from past experiences

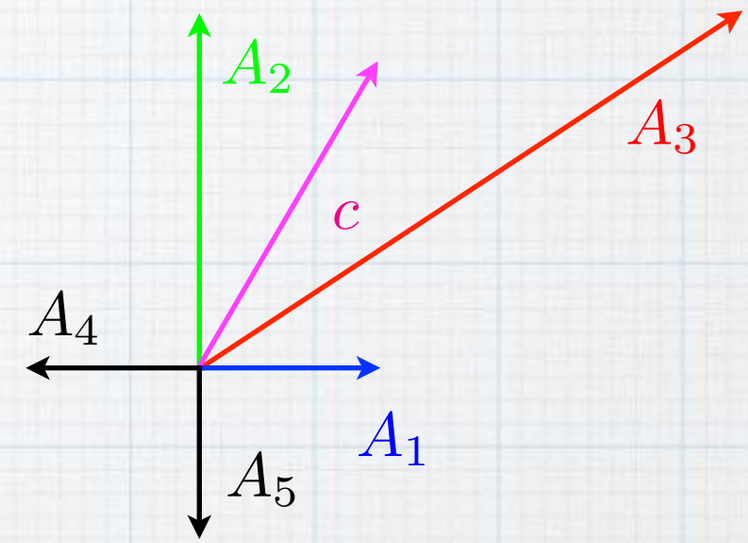
Read your travel book when we you reflected on

shortest paths and the rope problem
max flow - min cut

Now, can you give a better explanation of your findings?

Geometric interpretation





write c as linear non negative combination of A_i

P: $\max cx$

$$Ax \leq b$$

D: $\min yb$

$$yA = c$$

$$y \geq 0$$

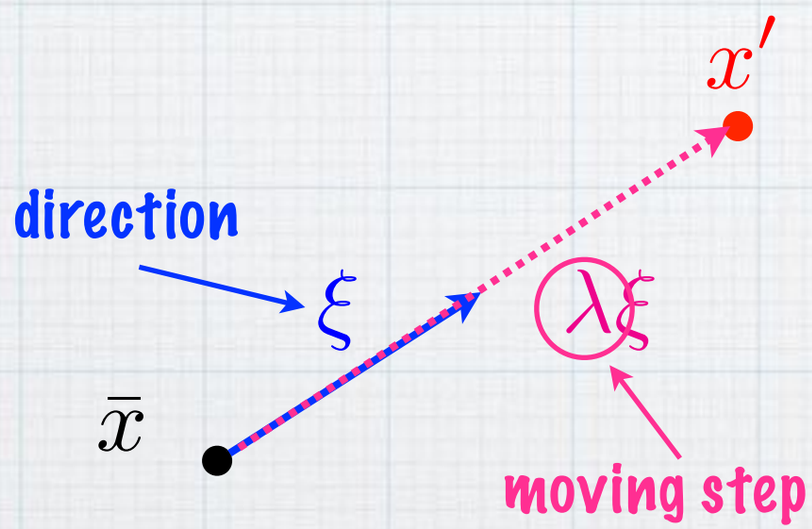
Can we improve a feasible solution?

P: $\max cx$

$$Ax \leq b$$

Let $\bar{x}$ be a feasible solution

Consider the new point $x' = \bar{x} + \lambda\xi$ $\qquad \lambda > 0$, $\xi \in \mathcal{R}^n$



x'

must be feasible $\qquad Ax' \leq b$

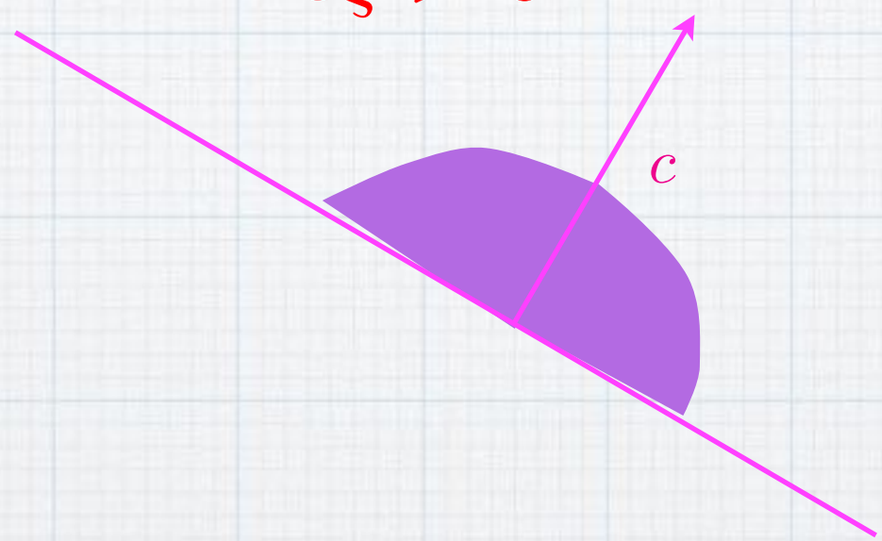
must improve the objective function value $\qquad cx' > c\bar{x}$

Growing feasible direction

What are the characteristics of the direction ξ ?

Improve the objective function

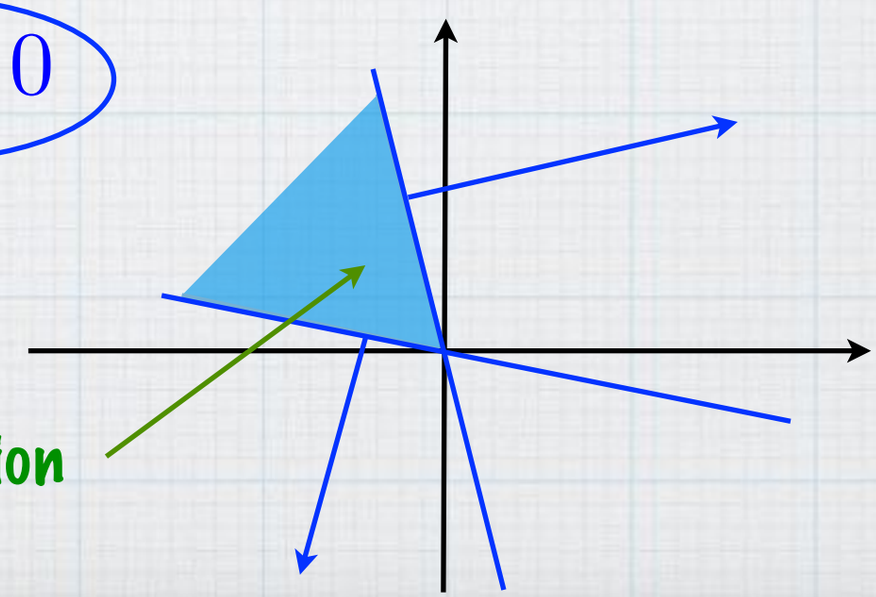
$$c\xi > 0$$



Allow to move (of a small step) without violating any constraint

$$A_I\xi \leq 0$$

polyhedral cone



any vector in this cone is a feasible growing direction

Optimality conditions

If a growing feasible direction ξ does not exist

$\bar{x}$ is optimum

Existence of a growing feasible direction

i) $c\xi > 0$
$A_I \xi \leq 0$

has a solution

c belongs to the cone generated by A_I

ii) $\eta A_I = c$
$\eta \geq 0$

has no solution

LP interpretation

RP: $\max c\xi$
$A_I \xi \leq 0$

Restricted Primal

is unbounded

RD: $\min 0\eta$
$\eta A_I = c$
$\eta \geq 0$

Restricted Dual

Optimality conditions

If a growing feasible direction ξ does not exist

$\bar{x}$ **is optimum**

Existence of a growing feasible direction

i) $$\begin{aligned} c\xi &> 0 \\ A_I\xi &\leq 0 \end{aligned}$$

has no solution

c belongs to the cone generated by A_I

ii) $$\begin{aligned} \eta A_I &= c \\ \eta &\geq 0 \end{aligned}$$

has a solution

LP interpretation

RP: $$\begin{aligned} \max\, & c\xi \\ & A_I\xi \leq 0 \end{aligned}$$

Restricted Primal

0 is an optimal solution

RD: $$\begin{aligned} \min\, & 0\eta \\ & \eta A_I = c \\ & \eta \geq 0 \end{aligned}$$

Restricted Dual

Either i) or ii) holds true but not both [Farkas Lemma]

Strong duality theorem

Consider

P: $\max cx$
$Ax \leq b$

D: $\min yb$
$yA = c$
$y \geq 0$

$\bar{x}$ and $\bar{y}$ are optimal solutions for P and D if and only if

$$c\bar{x} = \bar{y}b$$

Given a solution of ii) $\bar{\eta}$

construct a feasible solution for D

$$\bar{y}_i = \begin{cases} \bar{\eta}_i & i \in I \\ 0 & i \in \bar{I} \end{cases}$$

Determining the moving step

Assuming that a growing feasible direction ξ exists

What is the maximum value of λ such that x' is feasible?

Consider the non active constraints only

$$\bar{I} = \{i : A_i \bar{x} < b_i\}$$

distance from the constraint border

$$\lambda = \min\{\frac{b_i - A_i \bar{x}}{A_i \xi} : A_i \xi > 0, i \in \bar{I}\}$$

"approaching speed"

$$\lambda = \infty \text{ if } A_i \xi \leq 0, \forall i \in \bar{I}$$

the problem is unbounded

Sketch of a solution algorithm

Consider a feasible solution x;

while $\exists$ a growing feasible direction ξ

determine λ

move of λ along ξ

Choice of ξ

Among all possible directions select one
on the border of the polyhedron, if possible

Primal-Dual algorithm(A,b,c,x,unbounded,y)

optimal←false; unbounded←false;

$I \leftarrow \{i: A_i x = b_i\}$;

if $I=\emptyset$ then grow_along(c,x,I,unbounded);

while not optimal **and** not unbounded do

if $\{\eta A_I = c\}$ has no solution then compute $\xi : A_I \xi = 0,\ c\xi = 1$;

grow_along(ξ,x,I,unbounded);

else if $\{\eta A_I = c\}$ has a solution and $\exists h: \eta_h < 0$ then

compute $\xi : A_I \xi = -u_h$;

grow_along(ξ,x,I,unbounded);

else optimal←true; {the dual system has a solution: $\eta A_I = c,\ \eta \geq 0$}

$y_i \leftarrow \eta_i: i \in I,\ y_i \leftarrow 0$ otherwise.

grow_along(ξ ,x, I, unbounded)

$$\lambda \leftarrow \min\{\frac{b_i - A_i\bar{x}}{A_i\xi} : A_i\xi > 0, i \in \bar{I}, \infty\}$$

if $\lambda = \infty$ then unbounded $\leftarrow$ true

else $x \leftarrow x + \lambda\xi$

I $\leftarrow$ {i: $A_i x = b_i$}

Searching for a feasible solution

P: $\max cx$

$$Ax \leq b$$

If $b_i \geq 0 \quad i = 1, \ldots, n$ then $x = 0$ is feasible

$$I_+ = \{i : b_i \geq 0\}$$

$$I_- = \{i : b_i < 0\}$$

Construct an auxiliary problem

AP:

$$\max \quad -\sum_{i \in I_-} u_i$$

discourage the use of artificial variables

$x = 0, U_{I_-} = -b_{I_-}$ is feasible

$$\begin{aligned} A_{I_+} x \qquad\qquad &\leq b_{I_+} \\ A_{I_-} x \qquad -u_{I_-} &\leq b_{I_-} \\ -u_{I_-} &\leq 0 \end{aligned}$$

If the optimal solution $(\bar{x}, \bar{u}_{I_-})$ has $\bar{u}_{I_-} = 0$ then

$\bar{x}$ is feasible

otherwise P has no solution

Complementary slackness conditions

P: $\max cx$

$$Ax \leq b$$

D: $\min yb$

$$yA = c$$
$$y \geq 0$$

Given $\bar{x}$ and $\bar{y}$ feasible for **P** and **D**

the following three statements are equivalent

i) $\bar{x}$ and $\bar{y}$ are optimal

ii) $c\bar{x} = \bar{y}b$

iii) $\bar{y}(b - A\bar{x}) = 0$

if $\bar{y}_i \neq 0 \Rightarrow A_i\bar{x} = b_i$

if $A_i\bar{x} \neq b_i \Rightarrow \bar{y}_i = 0$

Complementary slackness conditions (2)

P:

$$\min cx$$
$$Ax \geq b$$
$$x \geq 0$$

D:

$$\max yb$$
$$yA \leq c$$
$$y \geq 0$$

$$\bar{y}(b - A\bar{x}) = 0$$
$$(c - \bar{y}A)\bar{x} = 0$$

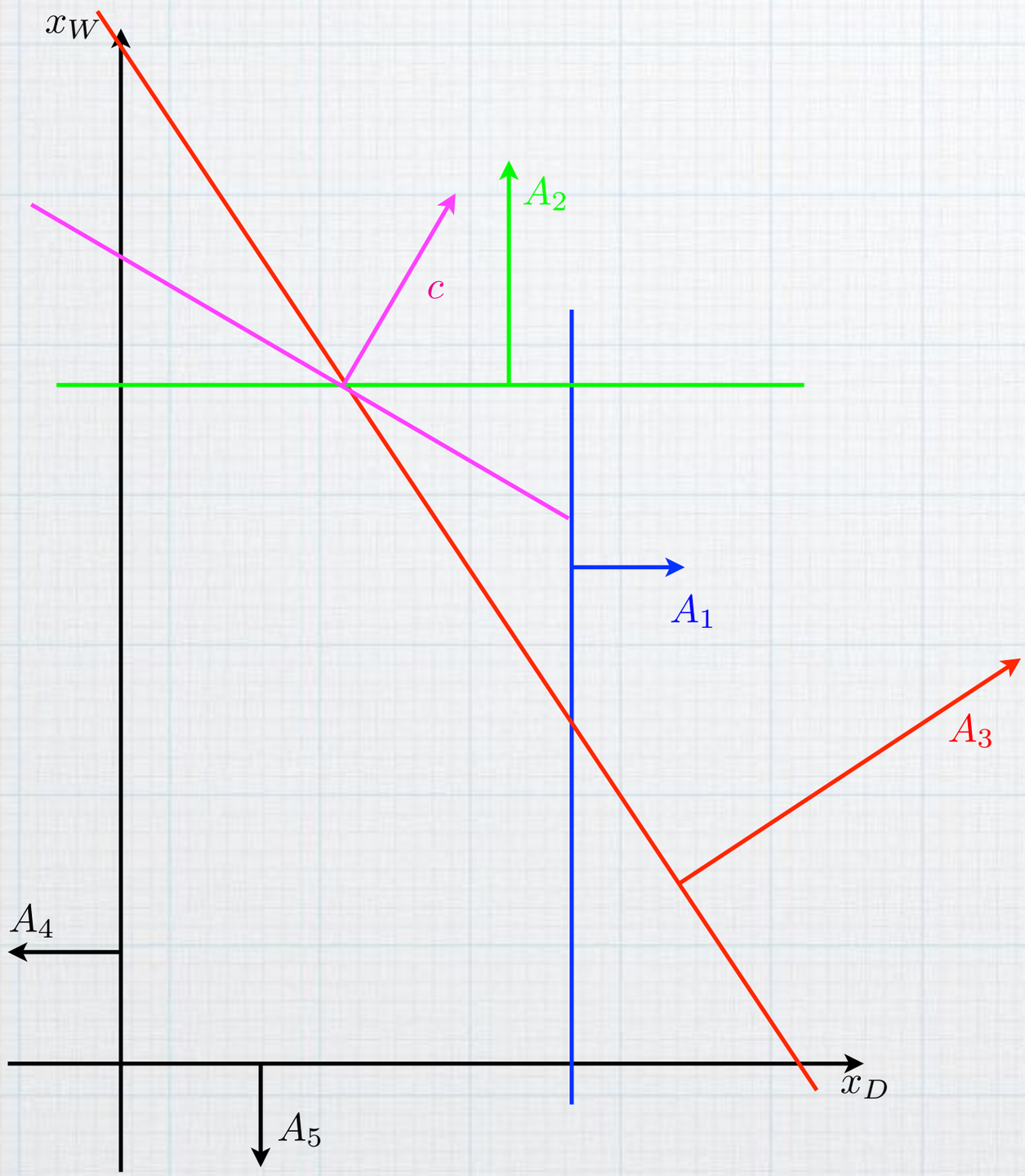
if $\bar{y}_i \neq 0$ $\Rightarrow$ $A_i\bar{x} = b_i$

if $A_i\bar{x} \neq b_i$ $\Rightarrow$ $\bar{y}_i = 0$

if $\bar{x}_j \neq 0$ $\Rightarrow$ $(c_j = \bar{y}A^j)$

if $(c_j \neq \bar{y}A^j)$ $\Rightarrow$ $\bar{x}_j = 0$

Economic interpretation of dual variables



$$\max 30x_D + 50x_W$$

$$\begin{aligned} x_D &\leq 4 \\ 2x_W &\leq 12 \\ 3x_D + 2x_W &\leq 18 \\ -x_D &\leq 0 \\ -x_W &\leq 0 \end{aligned}$$

$$\bar{x} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

$$\min 4y_1 + 12y_2 + 18y_3$$

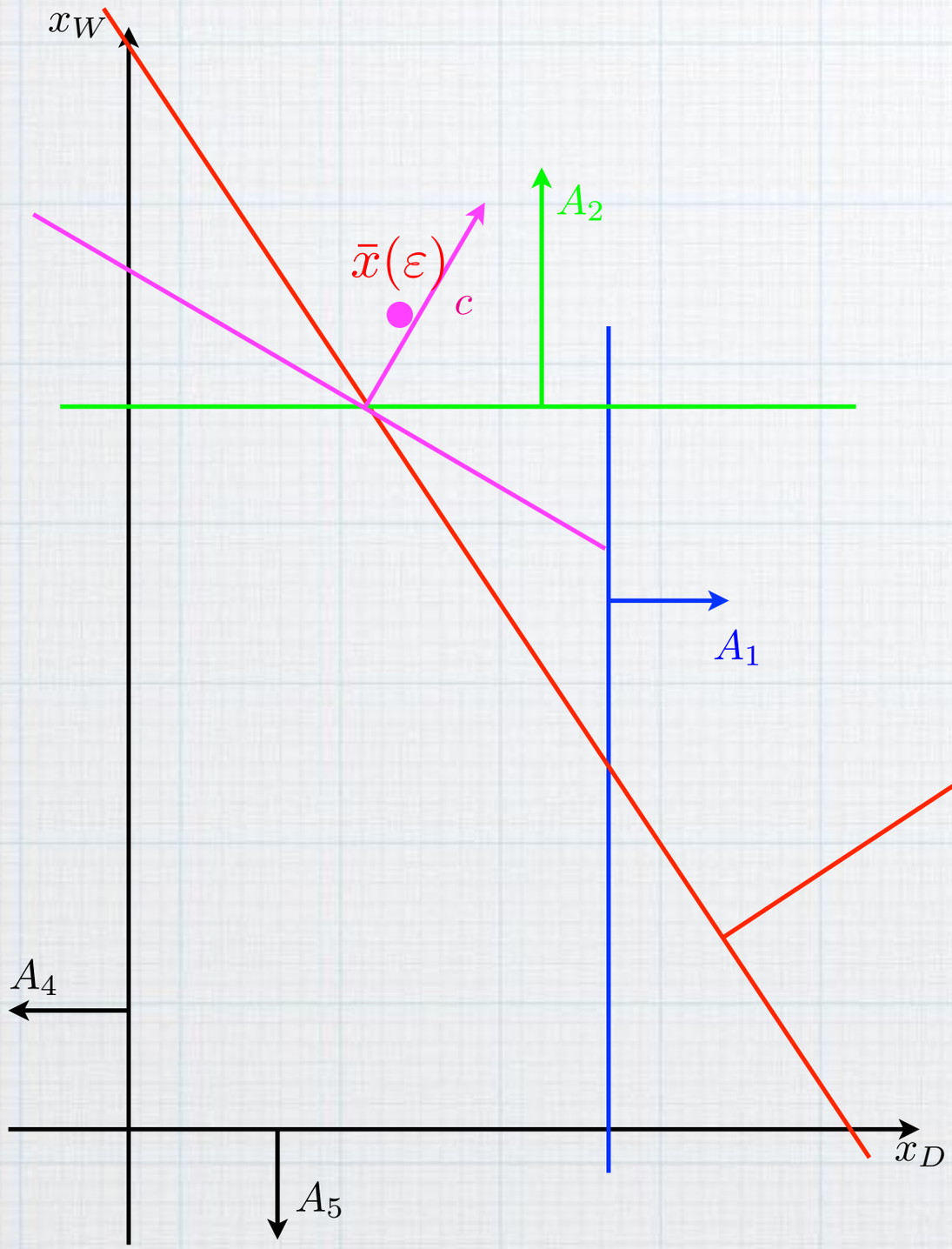
$$\begin{aligned} y_1 \quad\quad + 3y_3 - y_4 \quad\quad &= 30 \\ 2y_2 + 2y_3 \quad\quad - y_5 &= 50 \\ y_1, y_2, y_3, y_4, y_5 &\geq 0 \end{aligned}$$

$$\bar{y} = [0, 15, 10, 0, 0]$$

Are you willing to pay for one more unit of resource 1?

How much would you pay for one more unit of resource 2?

Economic interpretation of dual variables (2)



P(ε):

$$\max cx$$
$$A_1 x \le b_1 \quad +\varepsilon_1$$
$$A_2 x \le b_2 \quad +\varepsilon_2$$
$$\cdots$$
$$A_m x \le b_m \quad +\varepsilon_m$$

$\varepsilon_i, i = 1, \ldots, m$

parametric perturbations

optimal solution $\bar{x}$ $\quad \bar{x}(\varepsilon)$

D(ε):

$$\min b_1(y_1 + \epsilon_1) + b_2(y_2 + \epsilon_2) + \ldots + b_m(y_m + \epsilon_m)$$
$$y_1 A_1 + y_2 A_2 + \ldots + y_m A_m = c$$
$$y_1, y_2, \ldots, y_m \ge 0$$

optimal solution $\bar{y}$ **is still feasible**

by complementary slackness is also optimal

if ε_i are small enough $\bar{x}(\varepsilon)$ defines the same active constraints as $\bar{x}$

Shadow prices

$$c\bar{x}(\varepsilon) = \bar{y}(b + \varepsilon)$$
$$= c\bar{x} + \sum_{i=1}^{m} (\bar{y}_i \varepsilon_i)$$

impact on the objective function for each unit of perturbation

Changing cost coefficients

$$\begin{aligned} \max\ \alpha x_D + 50x_W \\ x_D \leq 4 \\ 2x_W \leq 12 \\ 3x_D + 2x_W \leq 18 \\ -x_D \leq 0 \\ -x_W \leq 0 \end{aligned}$$

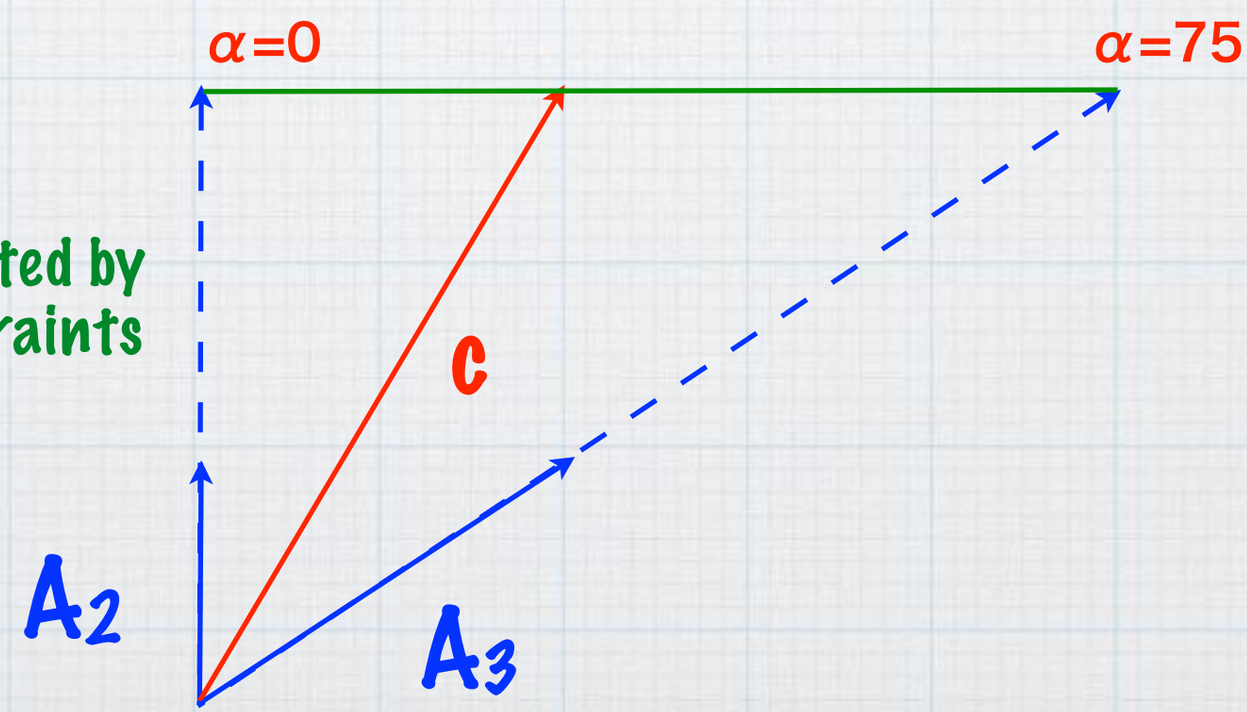
For which values of α $\bar{x}$ remains optimal?

$$\bar{x} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

The active constraints do not change

$$\begin{aligned} \min\ 4y_1 + 12y_2 + 18y_3 \\ y_1 + 3y_3 - y_4 = \alpha \\ 2y_2 + 2y_3 - y_5 = 50 \\ y_1, y_2, y_3, y_4, y_5 \geq 0 \end{aligned}$$

c remains within the cone generated by the gradients of the active constraints



Complementary bases and optimality conditions

$$\max \begin{bmatrix} c \end{bmatrix} \begin{bmatrix} x \end{bmatrix}$$

$$\begin{bmatrix} A_B \\ A \\ A_N \end{bmatrix} \begin{bmatrix} x \end{bmatrix} \leq \begin{bmatrix} b_B \\ b \\ b_N \end{bmatrix}$$

$$\min \begin{bmatrix} y_B & y & y_N \end{bmatrix} \begin{bmatrix} b_B \\ b \\ b_N \end{bmatrix}$$

$$\begin{bmatrix} y_B & y & y_N \end{bmatrix} \begin{bmatrix} A_B \\ A \\ A_N \end{bmatrix} = \begin{bmatrix} c \end{bmatrix}$$

$$y \geq 0$$

Partition of A, b, y

Basis B

n linearly independent rows of A

constraints holding as equalities

$$A_B x = b_B$$

variables equal to 0

$$y_N = 0$$

$$y_B A_B = c$$

Complementary bases and optimality conditions

$$\bar{x} = A_B^{-1} b_B$$

$$\bar{y}_B = c A_B^{-1} \qquad \bar{y}_N = 0$$

Primal feasibility

$$A_N \bar{x} \leq b_N$$

$$A_N A_B^{-1} b_B \leq b_N$$

$$\boxed{b_N - A_N A_B^{-1} b_B \geq 0}$$

Dual feasibility

$$\bar{y}_B \geq 0$$

$$\boxed{c A_B^{-1} \geq 0}$$

non negative reduced costs (of non basic variables in the dual)

What we have to spend to force a non basic variable into the basis

ILO of LP section

* write the dual of an LP problem
* understand the weak duality property
* understand the geometric interpretation of the primal and dual spaces
* understand the concept of growing feasible direction
* given a feasible point in the primal, ~~compute~~ write the conditions for the existence of a growing feasible direction, (if any) and a moving step, or certify the optimality
* write the auxiliary problem for finding a feasible solution
* write and understand the meaning of complementary slackness equations, use CS equations to check optimality
* perform some simple post optimality analyses
* interpret the complementary slackness and optimality conditions in terms of basic solutions

Resources Chapter 3

* Section 1.1 (just read section 1.2)
* Section 2 (2.1 included)
* Sections 3.1 and 3.2 (skip 3.3)
* Section 3.4 (skip 3.5)
* Section 4 (4.3 and 4.4 included)
* Section 5 (skip 5.2)